

## Training Configuration and Summary:

### Sequencing:

- timesteps: 30
- batch\_size: 128

### Model Hyperparameters:

- min\_units: 50
- max\_units: 150
- units\_step: 50
- dropout\_min: 0.2
- dropout\_max: 0.3
- dropout\_step: 0.1
- activation\_functions: ['tanh']
- optimizers: ['adam']

### Tuning:

- max\_epochs: 50
- hyperband\_iterations: 3

## LSTM Multi-Step Training Report

Report Generated: 2025-04-30 14:56:58

Training Start: 2025-04-23\_18-00-50

Training End: 2025-04-23\_18-07-37

Training Duration: 6.78 minutes

### Best Hyperparameters:

Number of Layers: 1

Activation: tanh

Optimizer: adam

Dropout Rate: 0.20

MAE: 1.0796

MSE: 1.9405

RMSE: 1.3930

R2 Score: 0.6790

### Multi-Step Forecast:

Forecast Steps: 20

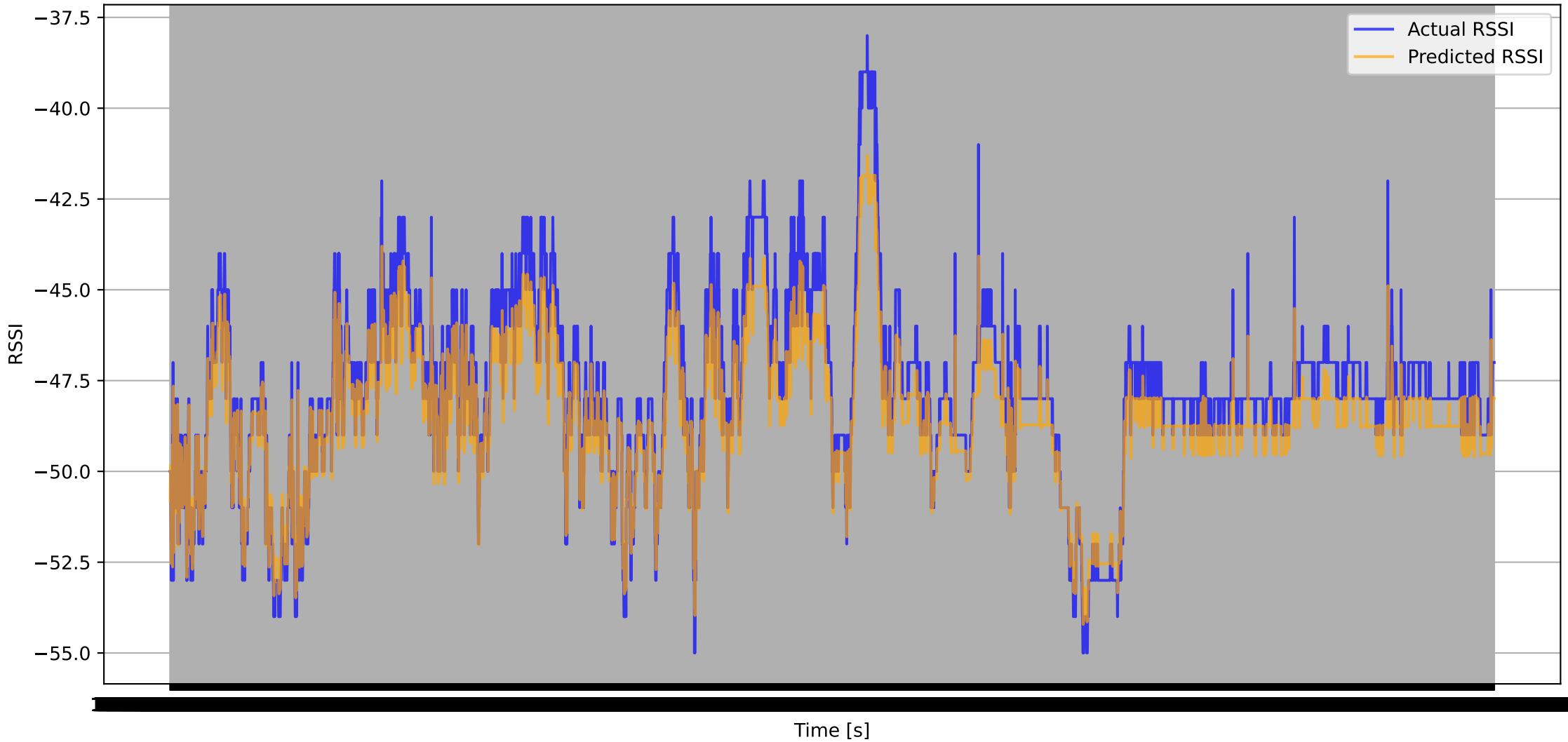
MAE: 0.0293

MSE: 0.0013

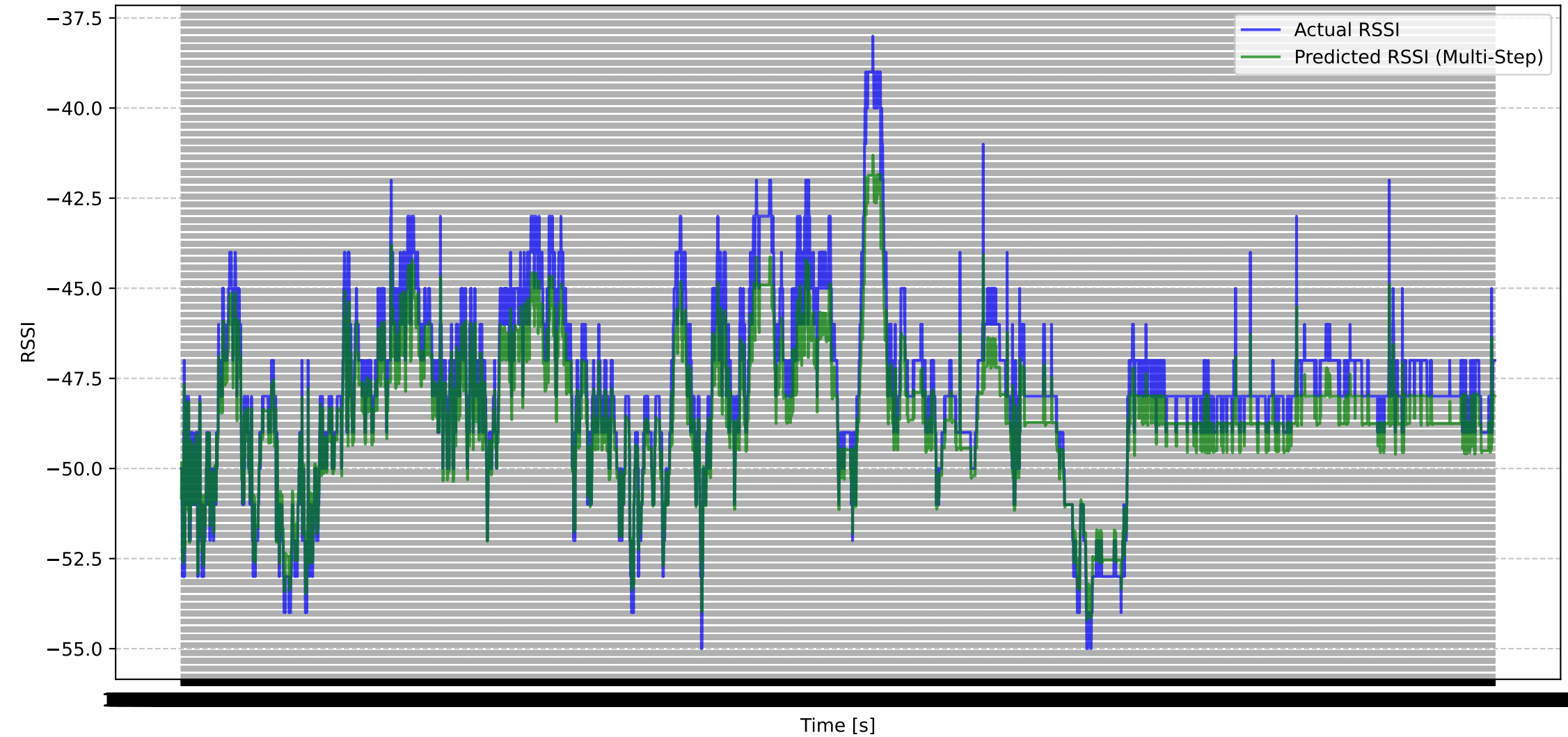
RMSE: 0.0360

R2 Score: 0.8206

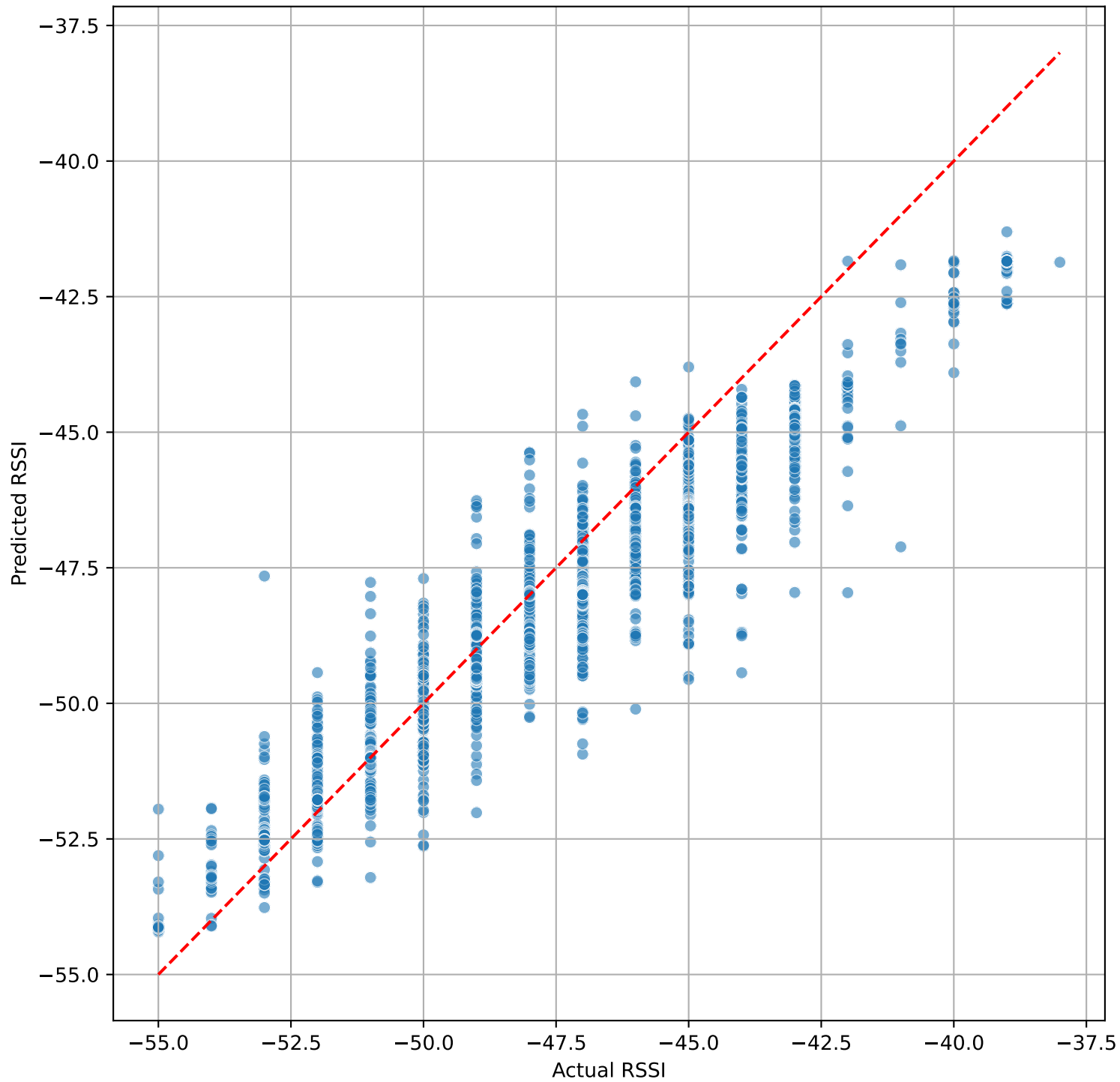
Actual vs Predicted RSSI (First Step)



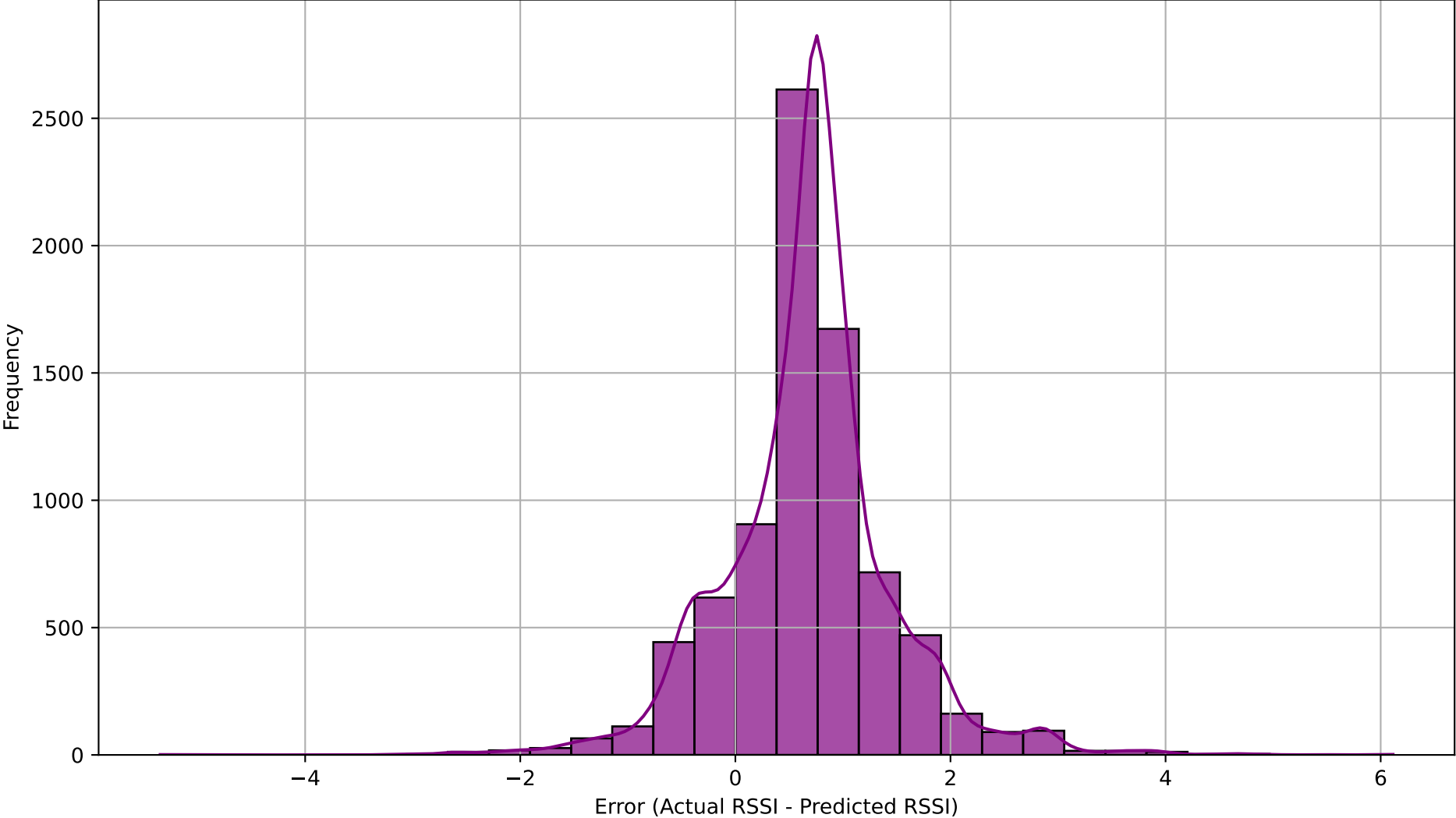
Actual vs Predicted RSSI (First Step vs Multi-Step)



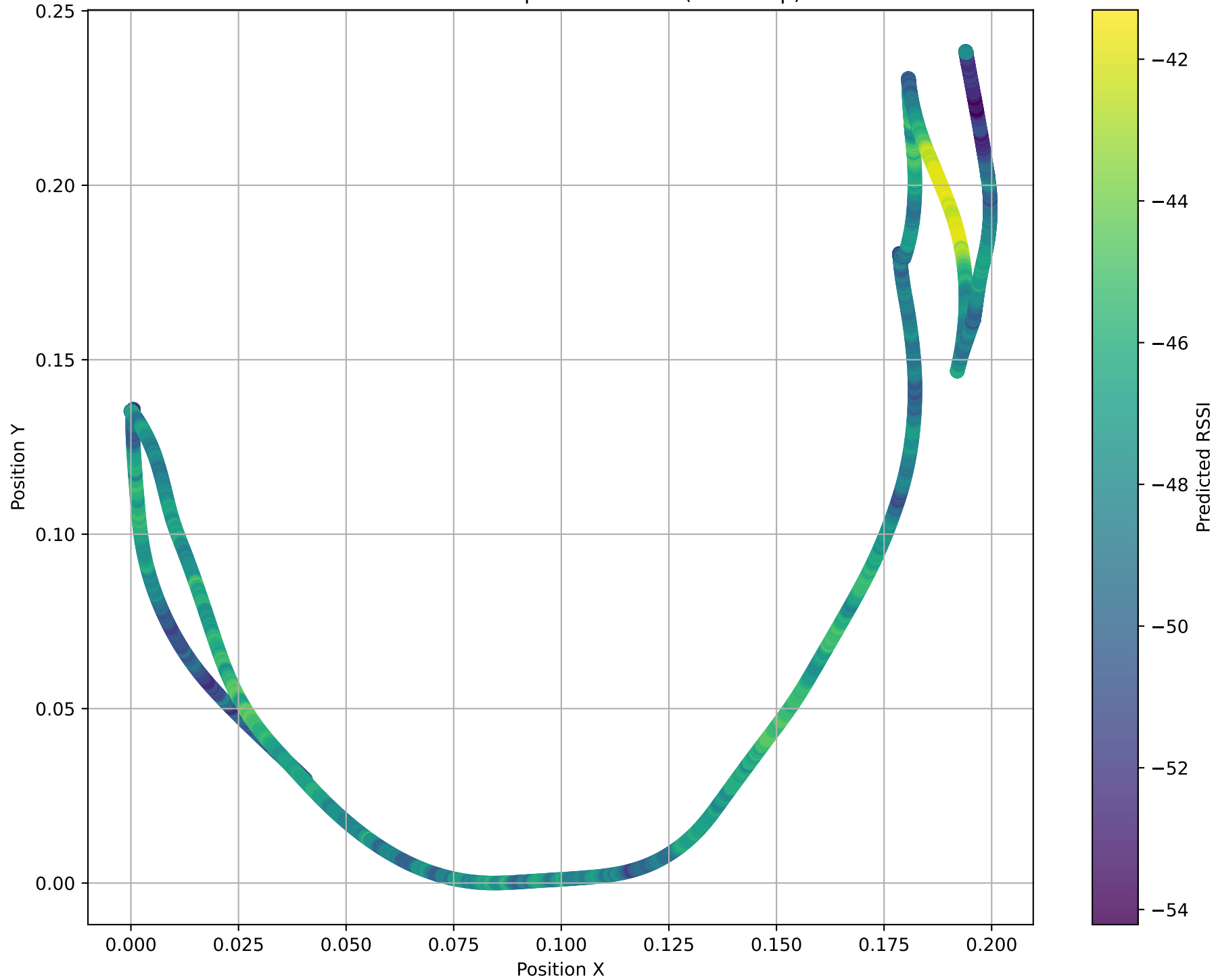
Actual vs Predicted RSSI (First Step)



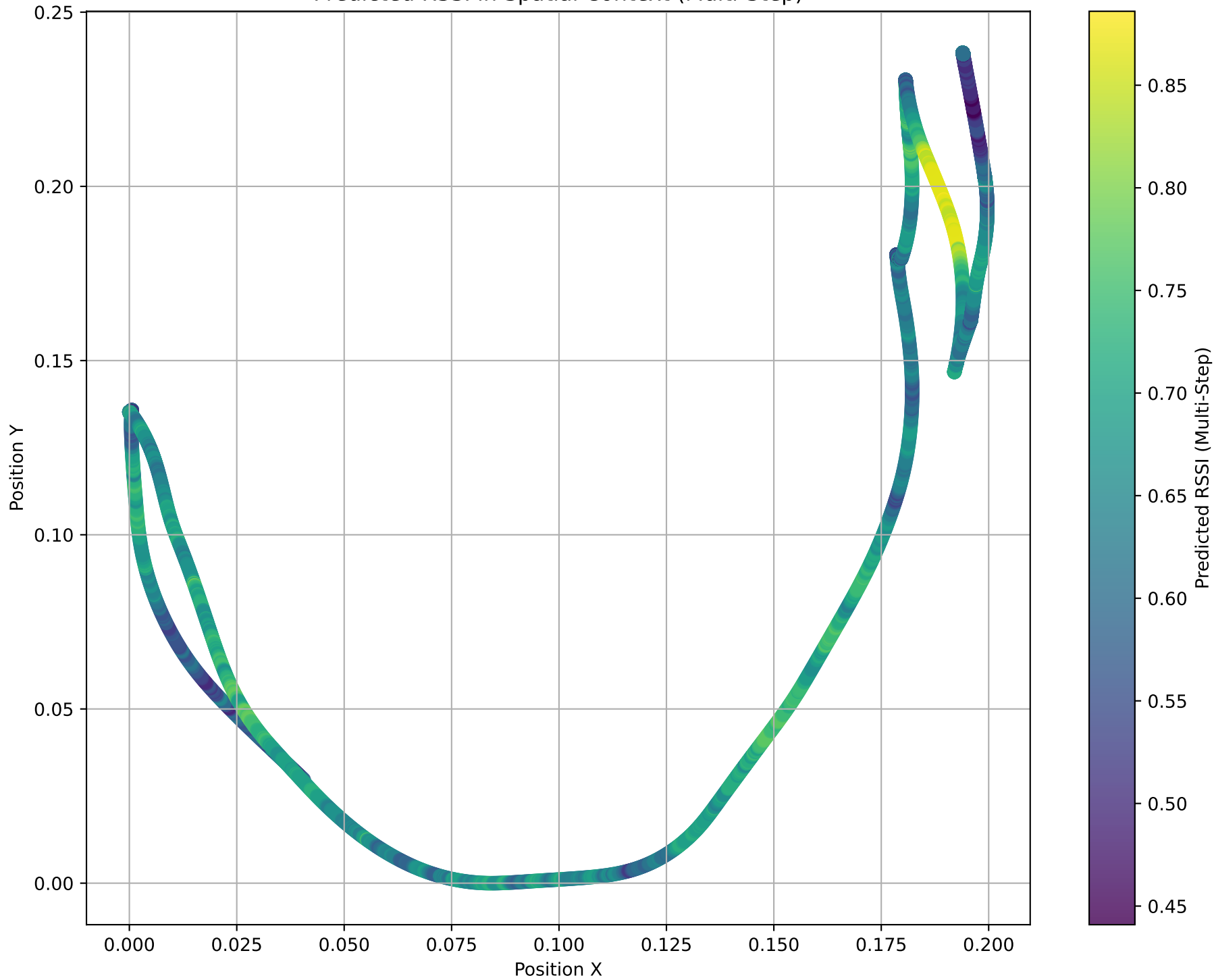
Distribution of Prediction Errors (First Step)



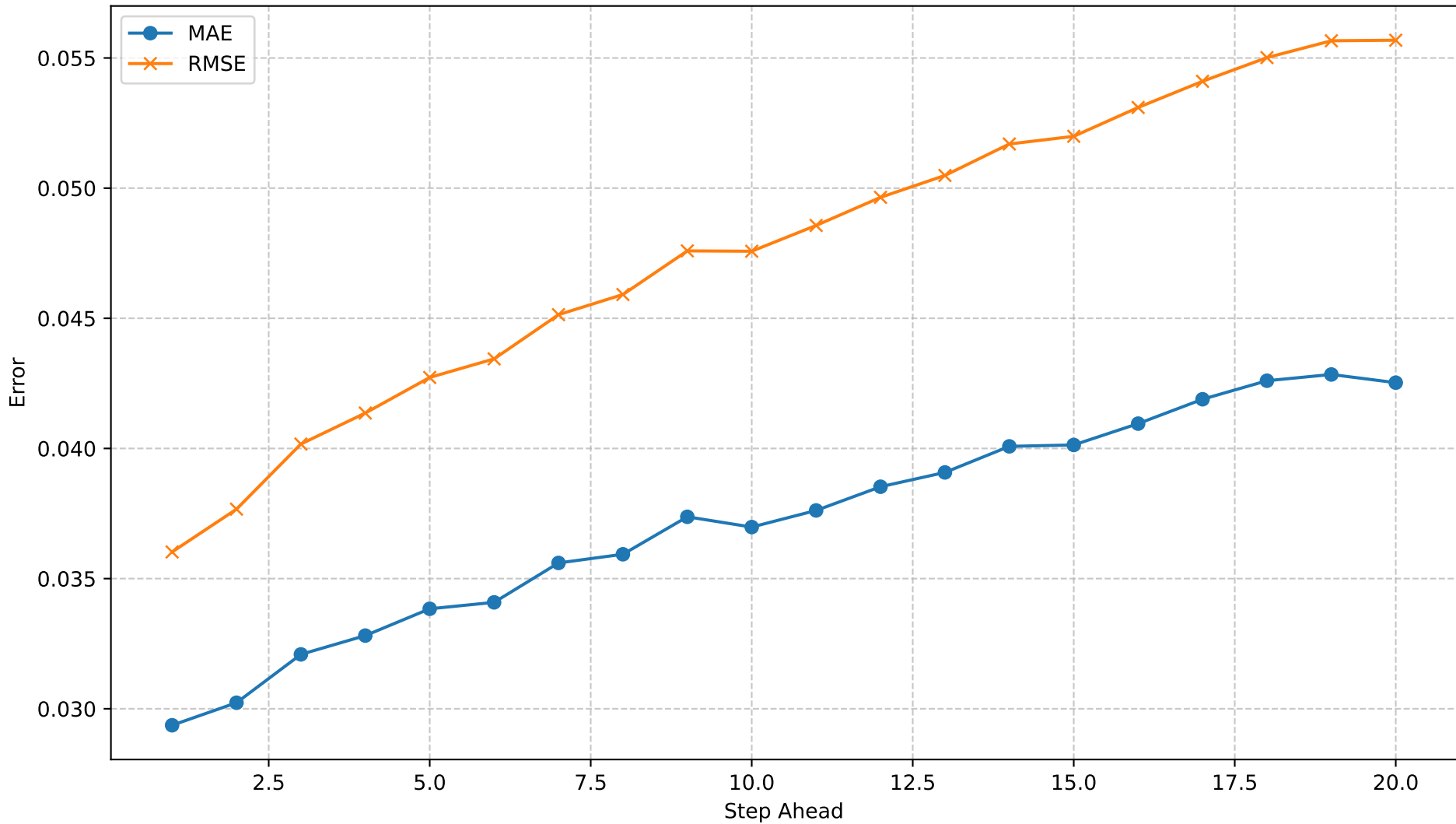
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step)



Multi-Step Forecast Error by Step





#### Resumo Automático:

- Forecast Steps configurados: 20
- Passos com MAE  $\leq 2.0$ : 20 passo(s)
- MAE no último passo: 0.0425

[ V ] O modelo mantém boa acurácia durante quase toda a sequência de previsões.